

Bonus Homework 3

Discrete Mathematics by Hongfei Fu, 2023.11.9

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1. (Bonus Homework 1) Prove that $[0, 1] \times [0, 1] \approx [0, 1]$.

Hint: Use Schröder-Berstein's Theorem and construct injections in two directions by decimal representation of real numbers.

Answer Area:

- First we construct a injection $f_1 : [0, 1] \times [0, 1] \rightarrow [0, 1]$:
For $(a_1, a_2) \in [0, 1] \times [0, 1]$, we proceed by considering the decimal representation of the real numbers, and denote

$$a_1 = 0.b_{1,1}b_{1,2}b_{1,3}b_{1,4} \dots$$

$$a_2 = 0.b_{2,1}b_{2,2}b_{2,3}b_{2,4} \dots$$

$$f_1((a_1, a_2)) = 0.b_{1,1}b_{2,1}b_{1,2}b_{2,2}b_{1,3}b_{2,3}b_{1,4}b_{2,4} \dots$$

Here we use $0 \dots b9999 \dots$ instead of $0 \dots (b+1)000 \dots$.

We can see that $f_1((a_1, a_2))$ is unique for certain (a_1, a_2) .

- Second, First we construct a injection $f_2 : [0, 1] \rightarrow [0, 1] \times [0, 1]$:
For $a_0 \in [0, 1]$, we also proceed by considering the decimal representation of the real numbers, and use $0 \dots b9999 \dots$ instead of $0 \dots (b+1)000 \dots$.
To begin with, we can build an injection $f_0 : [0, 1] \rightarrow (0, 1]$ by denoting:

$$f_0(x) = \begin{cases} 1 & x = 0 \\ \frac{1}{k+1} & x = \frac{1}{k} (k \in \mathbb{N}^*) \\ x & x \neq 0, \frac{1}{k} \notin \mathbb{N}^* \end{cases}$$

So we can transform the question into building an injection $f_2 : (0, 1] \rightarrow (0, 1] \times (0, 1]$

For an arbitrary $a \in (0, 1]$:

$$a = 0.b_1b_2b_3 \dots$$

Now we divide a into sections by cutting off behind b_n if $b_n \neq 0$. For example:

$$a = 0.10112008005 \dots \rightarrow 0.1 \ 01 \ 1 \ 2 \ 008 \ 005 \dots$$

Then we can denote this sections as $c_1, c_2, c_3, \dots$

$$a = 0.c_1c_2c_3 \dots$$

Now we can define:

$$f_2(a) = (0.c_1c_3c_5 \dots, 0.c_2c_4c_6 \dots)$$

We can see that f_2 is an injection from $(0, 1]$ to $(0, 1] \times (0, 1]$.

So there exists an injection from $[0, 1]$ to $[0, 1] \times [0, 1]$.

$\therefore$ According to Schröder-Berstein's Theorem, $[0, 1] \times [0, 1] \approx [0, 1]$.

2. (Bonus Homework 2) In this homework we examine the construction of natural numbers in set theory. First, for any set a , we define its *successor* a^+ as the set

$$a^+ := a \cup \{a\}.$$

By defining the successors, we have that

$$0 := \emptyset, 1 := \emptyset^+, 2 := (\emptyset^+)^+, 3 := ((\emptyset^+)^+)^+$$

We further say that a set A is *inductive* if $\emptyset \in A$ and $\forall a \in A (a^+ \in A)$. To define the set of natural numbers, we need the *Infinity Axiom* as follows:

Infinity Axiom: There exists an inductive set, i.e., $\exists A[\emptyset \in A \wedge (\forall a \in A (a^+ \in A))]$

Then we define the set of natural numbers as

$$\mathbb{N} := \{n \in A \mid n \in B \text{ for all inductive sets } B\}$$

where the set A is any inductive set whose existence is guaranteed by the Infinity Axiom. Prove that under the definition for natural numbers, the basic mathematical induction is correct. In detail, prove that given any predicate P , we have

$$\forall n \in \mathbb{N}(P(n)) \text{ iff } P(0) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1))$$

where $n+1$ here is equal to n^+ .

Answer Area:

- First we prove $\forall n \in \mathbb{N}(P(n)) \rightarrow P(0) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1))$:
 $\because$ A set A is *inductive* if $\emptyset \in A$ and $\forall a \in A (a^+ \in A)$, $0 := \emptyset$.
 $\therefore \forall B(B \text{ is an inductive set}), 0 \in B$.
 $\therefore 0 \in \mathbb{N}$.
 $\therefore P(0)$ is true.
 $\forall n \in \mathbb{N}(P(n))(n \in A \text{ and } n \in B \text{ for all inductive sets } B)$:
 $\because$ A set B is *inductive* if $\emptyset \in A$ and $\forall a \in A (a^+ \in A)$
 $\therefore n^+ \in A$ and $n^+ \in B$ for all *inductive* sets B .
 $\therefore P(n+1)$ is true.
 $\forall n \in \mathbb{N}(P(n) \rightarrow P(n+1))$.
 $\therefore \forall n \in \mathbb{N}(P(n)) \rightarrow P(0) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1))$
- Second, we prove $\forall n \in \mathbb{N}(P(n)) \leftarrow P(0) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1))$.
 From above we know that $0 \in \mathbb{N}$ and $P(0)$ is true, and if $n \in \mathbb{N}, n+1 \in \mathbb{N}$.
 $\therefore \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1))$.
 $\therefore P(0) \rightarrow P(1) \rightarrow \dots \rightarrow P(n) \rightarrow \dots$
 $\therefore \forall n \in \mathbb{N}(P(n))$.

$$\therefore \forall n \in \mathbb{N}(P(n)) \text{ iff } P(0) \wedge \forall n \in \mathbb{N}(P(n) \rightarrow P(n+1)).$$